

CURS #7

UNDE SUNTEM Multimi recursive, Multimi rec enumerable
reducere

A recursivă



χ_A calculabilă de
o M.T.

$$\chi_A(x) = \begin{cases} 1 & \text{dacă } x \in A \\ 0 & \text{dacă } x \notin A \end{cases}$$

A recursiv enumerabilă



f_A calculabilă de o M.T.

$$f_A(x) = \begin{cases} 1 & \text{dacă } x \in A \\ \uparrow & \text{dacă } x \notin A \end{cases}$$

Exp $HALT = \{ \langle i, x \rangle \mid M_i(x) \text{ se oprește} \}$
r.e. nu este recursivă

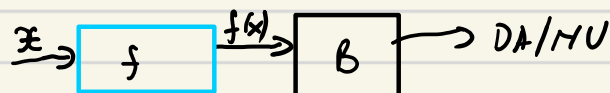
Exp2 $W = \{ p \in \mathbb{Z}[x] \mid \text{ecuația } p(x_1, \dots, x_n) = 0 \text{ are sol. întregi} \}$ r.e.

Teorema lui Matijasevic (curs 1) W nu este recursivă

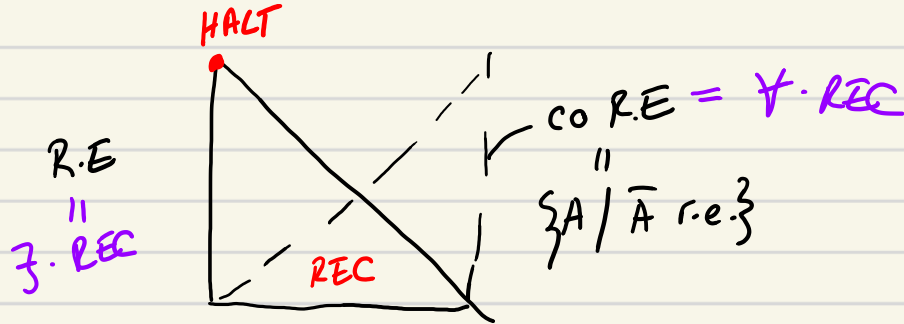
$A \leq_m B$ $\exists f: \mathbb{Z}^* \rightarrow \mathbb{Z}^*$ recursivă

a.i. $\forall x \in \mathbb{Z}^*$

$$x \in A \Leftrightarrow f(x) \in B$$



$$A \equiv_m B \Leftrightarrow A \leq_m B \text{ și } B \leq_m A$$



(T) $A \text{ r.e.} \Rightarrow A \leq_m HALT$

Dem $A \text{ r.e.} \Rightarrow \exists M_i$ $A = \{x \mid M_i(x) \downarrow\}$
 $f(x) = \langle i, x \rangle$ $x \in A \Leftrightarrow \langle i, x \rangle \in HALT$

$Tot = \{i \mid M_i(x) \text{ se opreste pt orice } x\}$

$HALT \leq_m TOT$ (dar $TOT \not\leq_m HALT$)

$\langle i, x \rangle \longrightarrow$ input z
 simulate $M_i(x)$
 decide $M_i(x) \downarrow$
 accept z

$HALT(i, x, t)$ $M_i(x)$ se opreste în t pași
 ↪ **recursiv!**

$HALT = \{\langle i, x \rangle \mid \exists t \text{ } HALT(i, x, t)\}$
 ↪ **recursiv**

(T) A este r.e.
 $\Uparrow$
 există un pred recursiv $P(x, y)$

$x \in A \Leftrightarrow \exists y \text{ } P(x, y)$

R.E. = $\exists \cdot REC$ **co-R.E. = $\forall \cdot REC$**

A este c.r.e. $\Rightarrow \bar{A}$ este r.e. există $P(x,y)$ recursiv

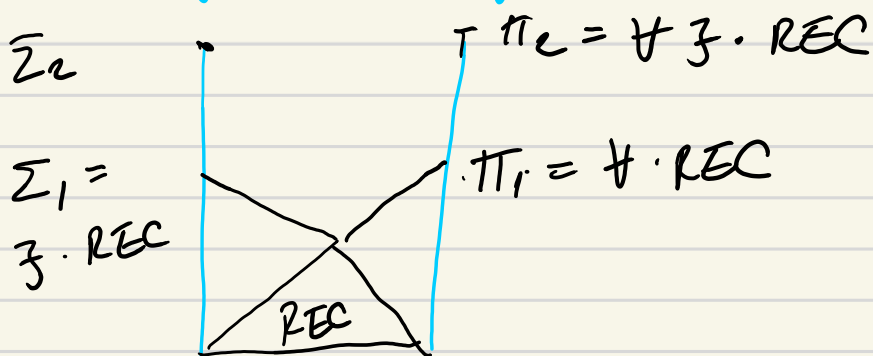
$$x \in \bar{A} \Leftrightarrow \exists y \ P(x,y)$$

$$x \in A \Leftrightarrow \forall y \neg P(x,y)$$

$$Tot = \{i \mid \forall x \ M_i(x) \text{ se oprește}\}$$

$$Tot = \{i \mid \forall x \exists t \underbrace{HALT(i,x,t)}_{rec.}\}$$

$$\exists x \exists y \Leftrightarrow \exists i = \langle x,y \rangle$$



$$\Sigma_n = \{A \text{ a.i. ex un pred rec } P$$

$$\text{a.i. } x \in A \Leftrightarrow \exists y_1 \forall y_2 \dots \exists y_n \ P(x, y_1, \dots, y_n)$$

$$\Pi_n = \{A \dots$$

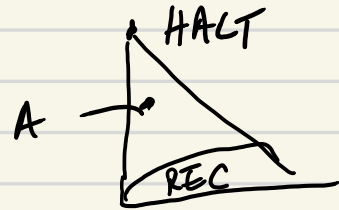
$$x \in A \Leftrightarrow \forall y_1 \exists y_2 \dots \forall y_n \ P(x, y_1, \dots, y_n)$$

Exp $Tot \in \Pi_2$ ie $Tot \Leftrightarrow \forall x \exists t \neg HALT(i, x, t)$

IERARHIA ARITMETICĂ

- $\Sigma_n \subset \Sigma_{n+1}$, $\Sigma_n \subset \Pi_{n+1}$
- Σ_n, Π_n au pb complete
- Exp TOT complete pt Π_2
- există A r.e, A nu este recursiv

$$A \leq_m \text{HALT}$$
$$\text{HALT} \not\leq_m A$$



Complexitate

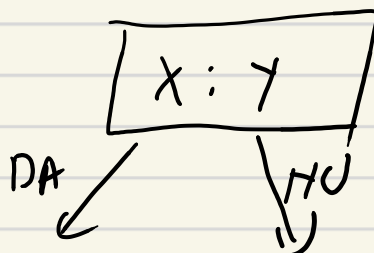
(Capitolul 2 Arora-Barak)

Problemele rezolvabile algoritmic eficient

Exp SORTAREA n numere $\rightarrow O(n \log n)$

MERGESORT
HEAPSORT
QUICKSORT

(T) Orice alg de sortare bazat pe comparații are complexitate $\Omega(n \log n)$



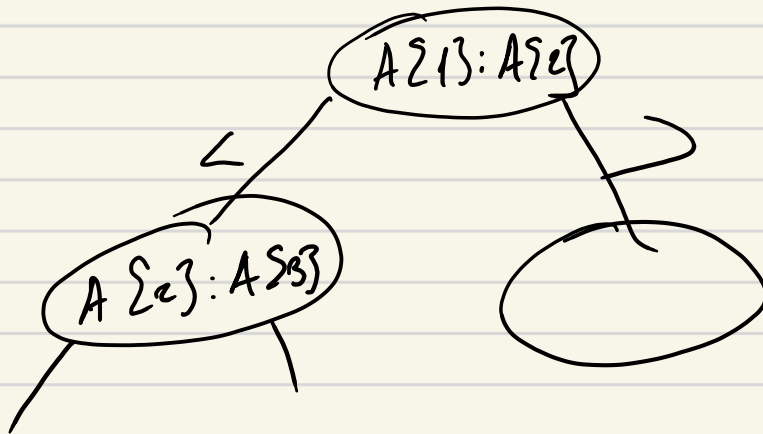
$$f = \Omega(n \log n) \quad \exists c > 0 \quad n_0 \geq 1$$

$$\forall n \geq n_0 \quad f(n) \geq c n \log(n)$$

Pt orice alg de sortare A

$$T_A(n) = \Omega(n \log(n))$$

Algoritme $\Rightarrow$ ARBORE de decizie
 INPUT ($A(1), \dots, A(n)$)



$$\boxed{T_{(2,1,3)}}$$

$$(A[2] < A[1] < A[3])$$

$$\underline{\text{height}(T) = T_p(n)}$$

Afirm Pt orice arbore de decizie pt ph. sortării
 $\text{height}(T) = \Omega(n \log(n))$

Obş T are $\geq n!$ frange.

Afirm Un arbore binar cu înălţime h are $\leq 2^h$ frange.

$$\forall T \quad h(T) \text{ verifică } 2^{h(T)} \geq n!$$

$$\forall T \quad h(T) \geq \log_2(n!)$$

FORMULA
LUI STIRLING

$$\lim_{n \rightarrow \infty} \frac{n!}{\left(\frac{n}{e}\right)^n \sqrt{2\pi n}} = 1$$

$$h(T) \geq \log_2(n!) \approx \log_2 \left(\left(\frac{n}{e}\right)^n \sqrt{2\pi n} \right)$$

$$\parallel$$
$$n \log_2(n/e) + \log_2(\sqrt{2\pi n})$$

$$\downarrow$$
$$O(n \log n)$$

REST Deopre ce n pot calcula eficient.